



**Generation Interconnection
System Impact Study Report
for
Queue Project AF1-030
“SANDWICH-PLANO 138 KV”
66.9 MW Capacity / 100 MW Energy**

August 2020

Table of Contents

1	Introduction.....	4
2	Preface.....	4
3	General	5
4	Point of Interconnection.....	5
5	Cost Summary	5
6	Transmission Owner Scope of Work	6
	Direct Connection Network Upgrades.....	6
	Non-Direct Connection Network Upgrades.....	7
7	Schedule.....	8
8	Transmission Owner Analysis.....	8
9	Interconnection Customer Requirements.....	8
10	Revenue Metering and SCADA Requirements	8
	10.1 PJM Requirements	8
	10.2 Interconnected Transmission Owner Requirements.....	8
11	Summer Peak Analysis	8
	11.1 Generation Deliverability	8
	11.2 Multiple Facility Contingency	9
	11.3 Contribution to Previously Identified Overloads.....	9
	11.4 Steady-State Voltage Requirements.....	9
	11.5 Potential Congestion due to Local Energy Deliverability.....	9
	11.6 System Reinforcements.....	11
	11.7 Flow Gate Details.....	12
	11.7.1 Index 1	12
	11.7.2 Index 2	14
	11.7.3 Index 3	17
	11.7.4 Index 4	20
	11.7.5 Index 5	23
	11.8 Queue Dependencies	26
	11.9 Contingency Descriptions.....	28
12	Light Load Analysis	31
13	Short Circuit Analysis.....	31

13.1	System Reinforcements - Short Circuit.....	31
14	Stability and Reactive Power	31
15	Affected Systems	32
15.1	MISO	32
16	Attachment 1: One Line Diagram	33

1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is ComEd.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

3 General

The Interconnection Customer (IC) has proposed a solar generating facility located in Kendall County, Illinois. The installed facilities will have a total capability of 100 MW with 66.9 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is December 01, 2022. This study does not imply a TO commitment to this in-service date.

Queue Number	AF1-030
Project Name	SANDWICH-PLANO 138 KV
State	Illinois
County	Kendall
Transmission Owner	ComEd
MFO	100
MWE	100
MWC	66.9
Fuel	Solar
Basecase Study Year	2023

A new service customer with a generating facility that could be commercially operable prior to June 1st of the basecase study year is required to request an interim deliverability analysis from PJM.

4 Point of Interconnection

Queue Position AF1-030, a 100 MW solar facility, proposes to interconnect with the ComEd transmission system by tying into 138kV Line 14609, 6.3 miles from TSS 146 Sandwich. Under an earlier queue AE2-341, a new Interconnection Substation would be built tying into 138kV line 14609, 6.3 miles from TSS 146 Sandwich. ComEd proposes to interconnect AF1-030 to the proposed Interconnection Substation, to be built by PJM queue position AE2-341.

5 Cost Summary

The AF1-030 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$3,000,000
Allocation towards System Network Upgrade Costs*	\$0
Total Costs	\$3,000,000

*As your project progresses through the study process and other projects modify their request or withdraw, then AF1-030 cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

Attachment Facilities

AF1-030 generator lead will interconnect to 138kV Interconnection Substation (see details in Direct Connection section below). The required Attachment Facilities are one 138kV line MOD, one dead-end structure and one revenue-metering as shown in the one-line diagram (See Attachment 1).

Scope of Work	Cost Estimate
Installation of one 138kV line MOD, one dead-end structure and one set of revenue metering (see notes below on cost estimate)	\$1,000,000
Total Cost Estimate (see notes below on cost estimate)	\$1,000,000

Direct Connection Network Upgrades

Prior to AF1-030 queue position, a new 138kV Interconnection Substation, would be built, approximately 6.3 miles from TSS 146 Sandwich, under an earlier queue position AE2-341 by looping in the 'Sandwich-Plano' 138kV line 14609.

In order to accommodate interconnection of AF1-030, the interconnection substation to be built by AE2-341, would need to be expanded to create a bus position.

The scope of work includes installation of one 138kV circuit breaker, to create a line position for AF1-030 generator lead, as shown in the one-line diagram below. It should be noted that if the AE2-341 project drops out, then scope of AF1-030 would change.

Scope of Work	Cost Estimate
Installation of one 138 kV circuit breaker at proposed future 138 kV substation	\$2,000,000
Total Cost Estimate (see notes below on cost estimate)	\$2,000,000

Non-Direct Connection Network Upgrades

None

Scope of Work	Cost Estimate
None	\$0
Total Cost Estimate (see notes below on cost estimate)	\$0

Notes on Cost Estimate:

- 1) These estimates are Order-of-Magnitude estimates of the costs that ComEd would bill to the customer for this interconnection. These estimates are based on a one-line electrical diagram of the project and the information provided by the Interconnection Customer.
- 2) There were no site visits performed for these estimates. There may be costs related to specific site related issues that are not identified in these estimates. The site reviews will be performed during the Facilities Study or during detailed engineering.
- 3) These estimates are not a guarantee of the maximum amount payable by the Interconnection Customer and the actual costs of ComEd's work may differ significantly from these estimates. Interconnection Customer will be responsible for paying actual costs of ComEd's work in accordance with Sections 212.1 and 217 of the PJM Open Access Transmission Tariff.
- 4) The Interconnection Customer is responsible for all engineering, procurement, testing and construction of all equipment on the Interconnection Customer's side of the POI.
- 5) These cost estimates do not include cost of acquiring right-of-way for the transmission line and purchasing any additional land, if needed, for the line terminations. The need and cost for acquiring property and associated legal costs will be investigation during Facilities Study for this project.

7 Schedule

The physical interconnection work scope schedule is dependent on prior queue project AE2-341 construction. See Section 11.6 for system network upgrade schedule.

8 Transmission Owner Analysis

See Section 6.

9 Interconnection Customer Requirements

The Interconnection Customer is responsible for all design and construction related activities on the Interconnection Customer's side of the Point of Interconnection.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 Interconnected Transmission Owner Requirements

The IC will be required to comply with all Interconnected Transmission Owner's revenue metering requirements for generation interconnection customers located at the following link:

ComEd interconnection requirements can be found at <https://www.pjm.com/planning/design-engineering/to-tech-standards/private-comed.aspx>

To the extent that these Applicable Technical Requirements and Standards may conflict with the terms and conditions of the Tariff, the Tariff shall control.

11 Summer Peak Analysis

The Queue Project AF1-030 was evaluated as a 100.0 MW (Capacity 66.9 MW) injection tapping the Sandwich to Plano West 138 kV line in the ComEd area. Project AF1-030 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-030 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
4409939	943120	AE2-341 TAP	138.0	CE	272803	WILTON ; R	138.0	CE	1	COMED_P4_08 3-38-BT3-4__	breaker	498.0	97.73	114.29	AC	88.03
4409940	943120	AE2-341 TAP	138.0	CE	272803	WILTON ; R	138.0	CE	1	COMED_P4_11 1-38-TR82__	breaker	449.0	82.82	100.36	AC	82.31

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
44099405	270926	WILTON ; B	345.0	CE	275232	WILTON ; 3M	345.0	CE	1	COMED_P4_11 2-65-BT5-6__	breaker	1379.0	115.36	115.52	AC	15.46
44099380	270927	WILTON ; R	345.0	CE	275233	WILTON ; 4M	345.0	CE	1	COMED_P4_11 2-65-BT2-3__	breaker	1379.0	117.77	117.94	AC	15.79
44099404	275232	WILTON ; 3M	345.0	CE	270644	WILTON ;	765.0	CE	1	COMED_P4_11 2-65-BT5-6__	breaker	1379.0	115.36	115.52	AC	15.46
44099381	275233	WILTON ; 4M	345.0	CE	270644	WILTON ;	765.0	CE	1	COMED_P4_11 2-65-BT2-3__	breaker	1379.0	117.77	117.94	AC	15.79

11.4 Steady-State Voltage Requirements

To be determined

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
45416574	272728	WATERMAN ; B	138.0	CE	271560	GLIDDEEN ; BT	138.0	CE	1	272803 WILTON ; R 138 943120 AE2-341 TAP 138 1	operation	321.0	116.6	133.41	AC	54.8

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS LOADIN G %	POST PROJECTIONS LOADIN G %	AC/D C	MW IMPACT
43372620	274804	UNIV PK N;RP	345.0	CE	243229	05OLIVE	345.0	AEP	1	COMED_P1-2_695_B2	operation	971.0	97.04	97.29	AC	10.4
44100130	943120	AE2-341 TAP	138.0	CE	272803	W PLANO ; R	138.0	CE	1	COMED_P1-2_138-L11106_B-R	operation	449.0	81.85	99.59	AC	82.6
44100131	943120	AE2-341 TAP	138.0	CE	272803	W PLANO ; R	138.0	CE	1	Base Case	operation	351.0	78.03	99.89	AC	79.05

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-030	Upgrade Number																																																																																																																												
44099404	4	WILTON ;3M 345.0 kV - WILTON ; 765.0 kV Ckt 1	ComEd N5145 : Build out the Wilton 765kV bus thereby allowing for 765kV L11216 (currently on Bus 6) to be relocated to Bus 8. Along with this line relocation, installation of 2-765kV BT CB's (6-8 & 8-2). This will eliminate the stuck breaker contingencies '112-65- BT4-5__' and '112-65-BT5-6__'. Project Type : CON Cost : \$12,000,000 Time Estimate : 30 Months Queue Project AF1-030 presently does not receive cost allocation for this upgrade. The potential cost allocation table is as follows:	\$12,000,000	\$0	N5145																																																																																																																												
44099405	2	WILTON ;B 345.0 kV - WILTON ;3M 345.0 kV Ckt 1																																																																																																																																
44099380	3	WILTON ;R 345.0 kV - WILTON ;4M 345.0 kV Ckt 1																																																																																																																																
44099381	5	WILTON ;4M 345.0 kV - WILTON ; 765.0 kV Ckt 1	<table><thead><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>\$ cost (\$12M)</th></tr></thead><tbody><tr><td>AD1-039</td><td>18.0</td><td>1.82%</td><td>\$ 0.218</td></tr><tr><td>AD1-098</td><td>14.9</td><td>1.50%</td><td>\$ 0.180</td></tr><tr><td>AD1-100</td><td>173.4</td><td>17.49%</td><td>\$ 2.099</td></tr><tr><td>AD1-133</td><td>47.1</td><td>4.75%</td><td>\$ 0.570</td></tr><tr><td>AD2-038</td><td>22.8</td><td>2.30%</td><td>\$ 0.276</td></tr><tr><td>AD2-047</td><td>34.3</td><td>3.46%</td><td>\$ 0.415</td></tr><tr><td>AD2-066</td><td>17.7</td><td>1.79%</td><td>\$ 0.215</td></tr><tr><td>AD2-102</td><td>29.7</td><td>2.99%</td><td>\$ 0.359</td></tr><tr><td>AD2-134</td><td>17.6</td><td>1.78%</td><td>\$ 0.213</td></tr><tr><td>AD2-194</td><td>19.7</td><td>1.98%</td><td>\$ 0.238</td></tr><tr><td>AE1-070</td><td>21.5</td><td>2.17%</td><td>\$ 0.260</td></tr><tr><td>AE1-113</td><td>44.5</td><td>4.49%</td><td>\$ 0.539</td></tr><tr><td>AE1-114</td><td>21.2</td><td>2.14%</td><td>\$ 0.257</td></tr><tr><td>AE1-163</td><td>51.7</td><td>5.21%</td><td>\$ 0.626</td></tr><tr><td>AE1-166</td><td>27.9</td><td>2.82%</td><td>\$ 0.338</td></tr><tr><td>AE1-172</td><td>45.9</td><td>4.63%</td><td>\$ 0.556</td></tr><tr><td>AE1-193</td><td>45.8</td><td>4.62%</td><td>\$ 0.555</td></tr><tr><td>AE1-194</td><td>45.8</td><td>4.62%</td><td>\$ 0.555</td></tr><tr><td>AE1-195</td><td>45.8</td><td>4.62%</td><td>\$ 0.555</td></tr><tr><td>AE1-205</td><td>29.7</td><td>2.99%</td><td>\$ 0.359</td></tr><tr><td>AE1-252</td><td>27.0</td><td>2.73%</td><td>\$ 0.327</td></tr><tr><td>AE2-107</td><td>15.2</td><td>1.53%</td><td>\$ 0.184</td></tr><tr><td>AE2-152</td><td>28.5</td><td>2.87%</td><td>\$ 0.344</td></tr><tr><td>AE2-153</td><td>35.1</td><td>3.55%</td><td>\$ 0.425</td></tr><tr><td>AE2-223</td><td>22.6</td><td>2.28%</td><td>\$ 0.274</td></tr><tr><td>AE2-255</td><td>15.1</td><td>1.53%</td><td>\$ 0.183</td></tr><tr><td>AE2-321</td><td>15.2</td><td>1.54%</td><td>\$ 0.184</td></tr><tr><td>AE2-341</td><td>23.8</td><td>2.40%</td><td>\$ 0.288</td></tr><tr><td>J974</td><td>17.9</td><td>1.80%</td><td>\$ 0.217</td></tr><tr><td>AF1-030</td><td>15.8</td><td>1.59%</td><td>\$ 0.191</td></tr></tbody></table>	Queue	MW contribution	Percentage of Cost	\$ cost (\$12M)	AD1-039	18.0	1.82%	\$ 0.218	AD1-098	14.9	1.50%	\$ 0.180	AD1-100	173.4	17.49%	\$ 2.099	AD1-133	47.1	4.75%	\$ 0.570	AD2-038	22.8	2.30%	\$ 0.276	AD2-047	34.3	3.46%	\$ 0.415	AD2-066	17.7	1.79%	\$ 0.215	AD2-102	29.7	2.99%	\$ 0.359	AD2-134	17.6	1.78%	\$ 0.213	AD2-194	19.7	1.98%	\$ 0.238	AE1-070	21.5	2.17%	\$ 0.260	AE1-113	44.5	4.49%	\$ 0.539	AE1-114	21.2	2.14%	\$ 0.257	AE1-163	51.7	5.21%	\$ 0.626	AE1-166	27.9	2.82%	\$ 0.338	AE1-172	45.9	4.63%	\$ 0.556	AE1-193	45.8	4.62%	\$ 0.555	AE1-194	45.8	4.62%	\$ 0.555	AE1-195	45.8	4.62%	\$ 0.555	AE1-205	29.7	2.99%	\$ 0.359	AE1-252	27.0	2.73%	\$ 0.327	AE2-107	15.2	1.53%	\$ 0.184	AE2-152	28.5	2.87%	\$ 0.344	AE2-153	35.1	3.55%	\$ 0.425	AE2-223	22.6	2.28%	\$ 0.274	AE2-255	15.1	1.53%	\$ 0.183	AE2-321	15.2	1.54%	\$ 0.184	AE2-341	23.8	2.40%	\$ 0.288	J974	17.9	1.80%	\$ 0.217	AF1-030	15.8	1.59%	\$ 0.191			
Queue	MW contribution	Percentage of Cost	\$ cost (\$12M)																																																																																																																															
AD1-039	18.0	1.82%	\$ 0.218																																																																																																																															
AD1-098	14.9	1.50%	\$ 0.180																																																																																																																															
AD1-100	173.4	17.49%	\$ 2.099																																																																																																																															
AD1-133	47.1	4.75%	\$ 0.570																																																																																																																															
AD2-038	22.8	2.30%	\$ 0.276																																																																																																																															
AD2-047	34.3	3.46%	\$ 0.415																																																																																																																															
AD2-066	17.7	1.79%	\$ 0.215																																																																																																																															
AD2-102	29.7	2.99%	\$ 0.359																																																																																																																															
AD2-134	17.6	1.78%	\$ 0.213																																																																																																																															
AD2-194	19.7	1.98%	\$ 0.238																																																																																																																															
AE1-070	21.5	2.17%	\$ 0.260																																																																																																																															
AE1-113	44.5	4.49%	\$ 0.539																																																																																																																															
AE1-114	21.2	2.14%	\$ 0.257																																																																																																																															
AE1-163	51.7	5.21%	\$ 0.626																																																																																																																															
AE1-166	27.9	2.82%	\$ 0.338																																																																																																																															
AE1-172	45.9	4.63%	\$ 0.556																																																																																																																															
AE1-193	45.8	4.62%	\$ 0.555																																																																																																																															
AE1-194	45.8	4.62%	\$ 0.555																																																																																																																															
AE1-195	45.8	4.62%	\$ 0.555																																																																																																																															
AE1-205	29.7	2.99%	\$ 0.359																																																																																																																															
AE1-252	27.0	2.73%	\$ 0.327																																																																																																																															
AE2-107	15.2	1.53%	\$ 0.184																																																																																																																															
AE2-152	28.5	2.87%	\$ 0.344																																																																																																																															
AE2-153	35.1	3.55%	\$ 0.425																																																																																																																															
AE2-223	22.6	2.28%	\$ 0.274																																																																																																																															
AE2-255	15.1	1.53%	\$ 0.183																																																																																																																															
AE2-321	15.2	1.54%	\$ 0.184																																																																																																																															
AE2-341	23.8	2.40%	\$ 0.288																																																																																																																															
J974	17.9	1.80%	\$ 0.217																																																																																																																															
AF1-030	15.8	1.59%	\$ 0.191																																																																																																																															

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-030	Upgrade Number
44099400,44099399	1	AE2-341 TAP 138.0 kV - W PLANO ; R 138.0 kV Ckt 1	<u>ComEd</u> ComEd 138kV L14609 ALDR rating is 573 MVA. No upgrade is required.	\$0	\$0	
			TOTAL COST	\$0	\$0	

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
44099399	943120	AE2-341 TAP	CE	272803	W PLANO ; R	CE	1	COMED_P4_083-38-BT3-4__	breaker	498.0	97.73	114.29	AC	88.03

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
272364	ESS H440N ;R	1.6549	50/50	1.6549
274850	MENDOTA H;RU	0.3844	50/50	0.3844
274855	GSG-6 ;RU	1.6188	50/50	1.6188
274872	LEE DEKAL;1U	4.5443	50/50	4.5443
290051	GSG-6; E	36.5590	50/50	36.5590
290108	LEEDK;1U E	106.9080	50/50	106.9080
916221	Z1-073 E	18.8638	50/50	18.8638
925301	AB2-191	0.6209	50/50	0.6209
925302	AB2-191 E	4.8411	50/50	4.8411
933431	AC2-156 C O1	6.1569	50/50	6.1569
933432	AC2-156 E O1	10.0455	50/50	10.0455
933911	AD1-013 C	6.5744	50/50	6.5744
933912	AD1-013 E	10.5020	50/50	10.5020
934431	AD1-067 C	0.4591	50/50	0.4591
934432	AD1-067 E	1.9302	50/50	1.9302
934701	AD1-098 C O1	23.0680	50/50	23.0680
934702	AD1-098 E O1	16.8420	50/50	16.8420
937001	AD2-134 C	9.5571	50/50	9.5571
937002	AD2-134 E	39.4804	50/50	39.4804
941131	AE2-107 C	33.4302	50/50	33.4302
941132	AE2-107 E	22.2868	50/50	22.2868
943121	AE2-341 C	88.5622	50/50	88.5622
943122	AE2-341 E	43.4888	50/50	43.4888
943381	AF1-009 C	1.9955	50/50	1.9955
943382	AF1-009 E	7.9820	50/50	7.9820
943591	AF1-030 C O1	58.8947	50/50	58.8947
943592	AF1-030 E O1	29.1393	50/50	29.1393
946671	AF1-331	8.5382	50/50	8.5382
WEC	WEC	0.1493	Confirmed LTF	0.1493
CBM-W2	CBM-W2	1.9083	Confirmed LTF	1.9083
NY	NY	0.0431	Confirmed LTF	0.0431
TVA	TVA	0.2030	Confirmed LTF	0.2030
O-066	O-066	0.4973	Confirmed LTF	0.4973
CBM-S2	CBM-S2	0.1040	Confirmed LTF	0.1040
CBM-S1	CBM-S1	0.9542	Confirmed LTF	0.9542
G-007	G-007	0.0770	Confirmed LTF	0.0770
MADISON	MADISON	1.8668	Confirmed LTF	1.8668
MEC	MEC	1.8019	Confirmed LTF	1.8019
GIBSON	GIBSON	0.0038	Confirmed LTF	0.0038
BLUEG	BLUEG	0.0746	Confirmed LTF	0.0746
TRIMBLE	TRIMBLE	0.0262	Confirmed LTF	0.0262
CBM-W1	CBM-W1	6.3926	Confirmed LTF	6.3926

11.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
44099405	270926	WILTON ;B	CE	275232	WILTON ;3M	CE	1	COMED_P4_112-65-BT5-6__	breaker	1379.0	115.36	115.52	AC	15.46

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
274771	LINCOLN ;2U	3.8691	50/50	3.8691
274772	LINCOLN ;3U	3.8691	50/50	3.8691
274773	LINCOLN ;4U	3.8691	50/50	3.8691
274774	LINCOLN ;5U	3.8691	50/50	3.8691
274775	LINCOLN ;6U	3.8691	50/50	3.8691
274776	LINCOLN ;7U	3.8691	50/50	3.8691
274777	LINCOLN ;8U	3.8691	50/50	3.8691
274830	U3-021 1	6.3090	Adder	7.42
274831	U3-021 2	6.3090	Adder	7.42
274881	PILOT HIL;1E	19.9563	Adder	23.48
274890	CAYUG;1U E	17.0197	Adder	20.02
274891	CAYUG;2U E	17.0197	Adder	20.02
275149	KELLYCK ;1E	19.9563	Adder	23.48
276167	Z1-106 E2	1.3075	Adder	1.54
276168	Z1-106 E1	1.3074	Adder	1.54
276169	Z1-107 E	2.6761	Adder	3.15
276170	Z1-108 E	2.5726	Adder	3.03
290021	O50 E	20.0450	Adder	23.58
290051	GSG-6; E	10.9017	Adder	12.83
290108	LEEDK;1U E	25.2161	Adder	29.67
293061	N-015 E	16.4567	Adder	19.36
293644	O22 E1	10.6010	Adder	12.47
293645	O22 E2	20.5784	Adder	24.21
294392	P-010 E	20.9000	Adder	24.59
910542	X3-005 E	0.7569	Adder	0.89
915011	Y3-013 1	3.8668	Adder	4.55
915021	Y3-013 2	3.8668	Adder	4.55
915031	Y3-013 3	3.8668	Adder	4.55
916221	Z1-073 E	5.6251	Adder	6.62
917502	Z2-087 E	21.5594	Adder	25.36
918052	AA1-018 E OP	16.9981	Adder	20.0
920272	AA2-123 E	2.5284	Adder	2.97
924041	AB2-047 C O1	4.0269	Adder	4.74
924042	AB2-047 E O1	26.9492	Adder	31.7
924471	AB2-096	43.8128	Adder	51.54
925302	AB2-191 E	1.4436	Adder	1.7
926311	AC1-109 1	1.9783	Adder	2.33
926321	AC1-109 2	1.9783	Adder	2.33
926331	AC1-110 1	1.9652	Adder	2.31
926341	AC1-110 2	1.9652	Adder	2.31
926351	AC1-111 1	0.7901	Adder	0.93
926361	AC1-111 2	0.7901	Adder	0.93

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
926371	AC1-111 3	0.7901	Adder	0.93
926381	AC1-111 4	0.7901	Adder	0.93
926391	AC1-111 5	0.7901	Adder	0.93
926401	AC1-111 6	0.7901	Adder	0.93
927451	AC1-142A 1	4.3292	Adder	5.09
927461	AC1-142A 2	4.3292	Adder	5.09
930481	AB1-089	68.0633	Adder	80.07
930501	AB1-091 O1	79.4049	Adder	93.42
930741	AB1-122 1O1	75.3653	Adder	88.67
930751	AB1-122 2O1	76.2988	Adder	89.76
932881	AC2-115 1	0.7443	Adder	0.88
932891	AC2-115 2	0.7443	Adder	0.88
932921	AC2-116	0.8619	Adder	1.01
933411	AC2-154 C	2.7084	Adder	3.19
933412	AC2-154 E	4.4189	Adder	5.2
933431	AC2-156 C O1	0.9902	Adder	1.16
933432	AC2-156 E O1	1.6156	Adder	1.9
933911	AD1-013 C	1.9249	Adder	2.26
933912	AD1-013 E	3.0748	Adder	3.62
933931	AD1-016 C	0.9608	Adder	1.13
933932	AD1-016 E	1.5676	Adder	1.84
934101	AD1-039 1	7.3858	Adder	8.69
934111	AD1-039 2	7.4773	Adder	8.8
934431	AD1-067 C	0.1369	Adder	0.16
934432	AD1-067 E	0.5756	Adder	0.68
934701	AD1-098 C O1	7.1725	Adder	8.44
934702	AD1-098 E O1	5.2367	Adder	6.16
934721	AD1-100 C	24.8701	Adder	29.26
934722	AD1-100 E	116.0607	Adder	136.54
934871	AD1-116 C	0.9899	Adder	1.16
934872	AD1-116 E	1.6151	Adder	1.9
934971	AD1-129 C	0.9353	Adder	1.1
934972	AD1-129 E	0.6236	Adder	0.73
935001	AD1-133 C O1	23.0433	Adder	27.11
935002	AD1-133 E O1	15.3622	Adder	18.07
936291	AD2-038 C O1	2.4322	Adder	2.86
936292	AD2-038 E O1	16.2771	Adder	19.15
936371	AD2-047 C O1	4.8465	Adder	5.7
936372	AD2-047 E O1	23.6625	Adder	27.84
936461	AD2-060	2.8509	Adder	3.35
936511	AD2-066 C O1	8.7722	Adder	10.32
936512	AD2-066 E O1	5.8482	Adder	6.88
937001	AD2-134 C	2.8499	Adder	3.35
937002	AD2-134 E	11.7728	Adder	13.85
937401	AD2-194 1	8.0764	Adder	9.5
937411	AD2-194 2	8.0764	Adder	9.5
938511	AE1-070 1	9.4897	Adder	11.16
938521	AE1-070 2	8.6821	Adder	10.21
938851	AE1-113 C	8.2686	Adder	9.73
938852	AE1-113 E	29.3159	Adder	34.49
939321	AE1-163 C O1	6.1117	Adder	7.19
939322	AE1-163 E O1	37.5434	Adder	44.17

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939351	AE1-166 C O1	12.2715	Adder	14.44
939352	AE1-166 E O1	11.3275	Adder	13.33
939401	AE1-172 C O1	6.7996	Adder	8.0
939402	AE1-172 E O1	31.9033	Adder	37.53
939741	AE1-205 C O1	10.4080	Adder	12.24
939742	AE1-205 E O1	14.3729	Adder	16.91
940101	AE1-252 C O1	13.6598	Adder	16.07
940102	AE1-252 E O1	9.1066	Adder	10.71
940752	AE2-062 E	0.1352	Adder	0.16
941131	AE2-107 C	7.5832	Adder	8.92
941132	AE2-107 E	5.0555	Adder	5.95
941551	AE2-152 C O1	14.1594	Adder	16.66
941552	AE2-152 E O1	9.4396	Adder	11.11
941561	AE2-153 C O1	5.1289	Adder	6.03
941562	AE2-153 E O1	24.0125	Adder	28.25
941731	AE2-173 O1	6.1952	Adder	7.29
942111	AE2-223 C	2.4161	Adder	2.84
942112	AE2-223 E	16.1695	Adder	19.02
942421	AE2-255 C O1	3.1320	Adder	3.68
942422	AE2-255 E O1	9.3961	Adder	11.05
942651	AE2-281 C O1	0.8731	Adder	1.03
942652	AE2-281 E O1	5.3633	Adder	6.31
942991	AE2-321 C	8.4588	Adder	9.95
942992	AE2-321 E	4.1663	Adder	4.9
943121	AE2-341 C	13.2198	Adder	15.55
943122	AE2-341 E	6.4917	Adder	7.64
943381	AF1-009 C	0.6205	Adder	0.73
943382	AF1-009 E	2.4818	Adder	2.92
943591	AF1-030 C O1	8.7913	Adder	10.34
943592	AF1-030 E O1	4.3497	Adder	5.12
943801	AF1-048 C	3.9542	Adder	4.65
943802	AF1-048 E	2.6361	Adder	3.1
944041	AF1-072	2.2279	Adder	2.62
944911	AF1-156 C	13.1136	Adder	15.43
944912	AF1-156 E	8.7424	Adder	10.29
945351	AF1-200 FTIR	322.9138	Merchant Transmission	322.9138
946661	AF1-330 C	2.0733	Adder	2.44
946662	AF1-330 E	0.4551	Adder	0.54
946671	AF1-331	2.4998	Adder	2.94
WEC	WEC	4.5647	Confirmed LTF	4.5647
CBM-W2	CBM-W2	25.2006	Confirmed LTF	25.2006
NY	NY	1.4251	Confirmed LTF	1.4251
TVA	TVA	1.6366	Confirmed LTF	1.6366
O-066	O-066	16.9008	Confirmed LTF	16.9008
CHEOAH	CHEOAH	0.0711	Confirmed LTF	0.0711
CBM-S1	CBM-S1	6.1514	Confirmed LTF	6.1514
G-007	G-007	2.6125	Confirmed LTF	2.6125
MADISON	MADISON	24.3210	Confirmed LTF	24.3210
MEC	MEC	16.0378	Confirmed LTF	16.0378
CALDERWOOD	CALDERWOOD	0.0567	Confirmed LTF	0.0567
BLUEG	BLUEG	2.2846	Confirmed LTF	2.2846
TRIMBLE	TRIMBLE	0.7897	Confirmed LTF	0.7897

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
CATAWBA	CATAWBA	0.3577	Confirmed LTF	0.3577
CBM-W1	CBM-W1	78.3626	Confirmed LTF	78.3626

11.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
44099380	270927	WILTON ; R	CE	275233	WILTON ;4M	CE	1	COMED_P4_112-65-BT2-3__	breaker	1379.0	117.77	117.94	AC	15.79

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
274771	LINCOLN ;2U	3.9737	50/50	3.9737
274772	LINCOLN ;3U	3.9737	50/50	3.9737
274773	LINCOLN ;4U	3.9737	50/50	3.9737
274774	LINCOLN ;5U	3.9737	50/50	3.9737
274775	LINCOLN ;6U	3.9737	50/50	3.9737
274776	LINCOLN ;7U	3.9737	50/50	3.9737
274777	LINCOLN ;8U	3.9737	50/50	3.9737
274830	U3-021 1	6.4436	Adder	7.58
274831	U3-021 2	6.4436	Adder	7.58
274881	PILOT HIL;1E	20.3728	Adder	23.97
274890	CAYUG;1U E	17.3594	Adder	20.42
274891	CAYUG;2U E	17.3594	Adder	20.42
275149	KELLYCK ;1E	20.3728	Adder	23.97
276167	Z1-106 E2	1.3354	Adder	1.57
276168	Z1-106 E1	1.3353	Adder	1.57
276169	Z1-107 E	2.7333	Adder	3.22
276170	Z1-108 E	2.6275	Adder	3.09
290021	O50 E	20.4721	Adder	24.08
290051	GSG-6; E	11.1333	Adder	13.1
290108	LEEDK;1U E	25.7533	Adder	30.3
293061	N-015 E	16.8004	Adder	19.77
293644	O22 E1	10.8250	Adder	12.74
293645	O22 E2	21.0133	Adder	24.72
294392	P-010 E	21.3365	Adder	25.1
910542	X3-005 E	0.7734	Adder	0.91
915011	Y3-013 1	3.9494	Adder	4.65
915021	Y3-013 2	3.9494	Adder	4.65
915031	Y3-013 3	3.9494	Adder	4.65
916221	Z1-073 E	5.7446	Adder	6.76
917502	Z2-087 E	21.9972	Adder	25.88
918052	AA1-018 E OP	17.3697	Adder	20.43
920272	AA2-123 E	2.5823	Adder	3.04
924041	AB2-047 C O1	4.1087	Adder	4.83
924042	AB2-047 E O1	27.4965	Adder	32.35
924471	AB2-096	44.7470	Adder	52.64
925302	AB2-191 E	1.4743	Adder	1.73
926311	AC1-109 1	2.0204	Adder	2.38

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
926321	AC1-109 2	2.0204	Adder	2.38
926331	AC1-110 1	2.0070	Adder	2.36
926341	AC1-110 2	2.0070	Adder	2.36
926351	AC1-111 1	0.8070	Adder	0.95
926361	AC1-111 2	0.8070	Adder	0.95
926371	AC1-111 3	0.8070	Adder	0.95
926381	AC1-111 4	0.8070	Adder	0.95
926391	AC1-111 5	0.8070	Adder	0.95
926401	AC1-111 6	0.8070	Adder	0.95
927451	AC1-142A 1	4.4224	Adder	5.2
927461	AC1-142A 2	4.4224	Adder	5.2
930481	AB1-089	69.5172	Adder	81.78
930501	AB1-091 O1	81.0552	Adder	95.36
930741	AB1-122 1O1	76.9684	Adder	90.55
930751	AB1-122 2O1	77.9263	Adder	91.68
932881	AC2-115 1	0.7601	Adder	0.89
932891	AC2-115 2	0.7601	Adder	0.89
932921	AC2-116	0.8802	Adder	1.04
933411	AC2-154 C	2.7647	Adder	3.25
933412	AC2-154 E	4.5109	Adder	5.31
933431	AC2-156 C O1	1.0112	Adder	1.19
933432	AC2-156 E O1	1.6499	Adder	1.94
933911	AD1-013 C	1.9659	Adder	2.31
933912	AD1-013 E	3.1403	Adder	3.69
933931	AD1-016 C	0.9813	Adder	1.15
933932	AD1-016 E	1.6010	Adder	1.88
934101	AD1-039 1	7.5429	Adder	8.87
934111	AD1-039 2	7.6368	Adder	8.98
934431	AD1-067 C	0.1398	Adder	0.16
934432	AD1-067 E	0.5878	Adder	0.69
934701	AD1-098 C O1	7.3253	Adder	8.62
934702	AD1-098 E O1	5.3482	Adder	6.29
934721	AD1-100 C	25.3636	Adder	29.84
934722	AD1-100 E	118.3633	Adder	139.25
934871	AD1-116 C	1.0116	Adder	1.19
934872	AD1-116 E	1.6505	Adder	1.94
934971	AD1-129 C	0.9552	Adder	1.12
934972	AD1-129 E	0.6368	Adder	0.75
935001	AD1-133 C O1	23.5222	Adder	27.67
935002	AD1-133 E O1	15.6815	Adder	18.45
936291	AD2-038 C O1	2.4848	Adder	2.92
936292	AD2-038 E O1	16.6288	Adder	19.56
936371	AD2-047 C O1	4.9474	Adder	5.82
936372	AD2-047 E O1	24.1549	Adder	28.42
936461	AD2-060	2.9102	Adder	3.42
936511	AD2-066 C O1	8.9580	Adder	10.54
936512	AD2-066 E O1	5.9720	Adder	7.03
937001	AD2-134 C	2.9104	Adder	3.42
937002	AD2-134 E	12.0229	Adder	14.14
937401	AD2-194 1	8.2498	Adder	9.71
937411	AD2-194 2	8.2493	Adder	9.71
938511	AE1-070 1	9.6935	Adder	11.4

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
938521	AE1-070 2	8.8679	Adder	10.43
938851	AE1-113 C	8.4447	Adder	9.93
938852	AE1-113 E	29.9404	Adder	35.22
939321	AE1-163 C O1	6.2438	Adder	7.35
939322	AE1-163 E O1	38.3545	Adder	45.12
939351	AE1-166 C O1	12.5188	Adder	14.73
939352	AE1-166 E O1	11.5558	Adder	13.6
939401	AE1-172 C O1	6.9344	Adder	8.16
939402	AE1-172 E O1	32.5358	Adder	38.28
939741	AE1-205 C O1	10.6193	Adder	12.49
939742	AE1-205 E O1	14.6648	Adder	17.25
940101	AE1-252 C O1	13.9306	Adder	16.39
940102	AE1-252 E O1	9.2871	Adder	10.93
940752	AE2-062 E	0.1381	Adder	0.16
941131	AE2-107 C	7.7444	Adder	9.11
941132	AE2-107 E	5.1629	Adder	6.07
941551	AE2-152 C O1	14.4447	Adder	16.99
941552	AE2-152 E O1	9.6298	Adder	11.33
941561	AE2-153 C O1	5.2345	Adder	6.16
941562	AE2-153 E O1	24.5070	Adder	28.83
941731	AE2-173 O1	6.3210	Adder	7.44
942111	AE2-223 C	2.4652	Adder	2.9
942112	AE2-223 E	16.4979	Adder	19.41
942421	AE2-255 C O1	3.1988	Adder	3.76
942422	AE2-255 E O1	9.5963	Adder	11.29
942651	AE2-281 C O1	0.8920	Adder	1.05
942652	AE2-281 E O1	5.4792	Adder	6.45
942991	AE2-321 C	8.6393	Adder	10.16
942992	AE2-321 E	4.2552	Adder	5.01
943121	AE2-341 C	13.5012	Adder	15.88
943122	AE2-341 E	6.6298	Adder	7.8
943381	AF1-009 C	0.6337	Adder	0.75
943382	AF1-009 E	2.5347	Adder	2.98
943591	AF1-030 C O1	8.9784	Adder	10.56
943592	AF1-030 E O1	4.4422	Adder	5.23
943801	AF1-048 C	4.0386	Adder	4.75
943802	AF1-048 E	2.6924	Adder	3.17
944041	AF1-072	2.2755	Adder	2.68
944911	AF1-156 C	13.3837	Adder	15.75
944912	AF1-156 E	8.9225	Adder	10.5
945351	AF1-200 FTIR	329.7310	Merchant Transmission	329.7310
946661	AF1-330 C	2.1175	Adder	2.49
946662	AF1-330 E	0.4648	Adder	0.55
946671	AF1-331	2.5531	Adder	3.0
WEC	WEC	4.6620	Confirmed LTF	4.6620
CBM-W2	CBM-W2	25.7248	Confirmed LTF	25.7248
NY	NY	1.4555	Confirmed LTF	1.4555
TVA	TVA	1.6688	Confirmed LTF	1.6688
O-066	O-066	17.2637	Confirmed LTF	17.2637
CHEOAH	CHEOAH	0.0731	Confirmed LTF	0.0731
CBM-S1	CBM-S1	6.2707	Confirmed LTF	6.2707
G-007	G-007	2.6686	Confirmed LTF	2.6686

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
MADISON	MADISON	24.8371	Confirmed LTF	24.8371
MEC	MEC	16.3762	Confirmed LTF	16.3762
CALDERWOOD	CALDERWOOD	0.0586	Confirmed LTF	0.0586
BLUEG	BLUEG	2.3349	Confirmed LTF	2.3349
TRIMBLE	TRIMBLE	0.8069	Confirmed LTF	0.8069
CATAWBA	CATAWBA	0.3654	Confirmed LTF	0.3654
CBM-W1	CBM-W1	80.0390	Confirmed LTF	80.0390

11.7.4 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
44099404	275232	WILTON ;3M	CE	270644	WILTON ;	CE	1	COMED_P4_112-65-BT5-6__	breaker	1379.0	115.36	115.52	AC	15.46

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
274771	LINCOLN ;2U	3.8691	50/50	3.8691
274772	LINCOLN ;3U	3.8691	50/50	3.8691
274773	LINCOLN ;4U	3.8691	50/50	3.8691
274774	LINCOLN ;5U	3.8691	50/50	3.8691
274775	LINCOLN ;6U	3.8691	50/50	3.8691
274776	LINCOLN ;7U	3.8691	50/50	3.8691
274777	LINCOLN ;8U	3.8691	50/50	3.8691
274830	U3-021 1	6.3090	Adder	7.42
274831	U3-021 2	6.3090	Adder	7.42
274881	PILOT HIL;1E	19.9563	Adder	23.48
274890	CAYUG;1U E	17.0197	Adder	20.02
274891	CAYUG;2U E	17.0197	Adder	20.02
275149	KELLYCK ;1E	19.9563	Adder	23.48
276167	Z1-106 E2	1.3075	Adder	1.54
276168	Z1-106 E1	1.3074	Adder	1.54
276169	Z1-107 E	2.6761	Adder	3.15
276170	Z1-108 E	2.5726	Adder	3.03
290021	O50 E	20.0450	Adder	23.58
290051	GSG-6; E	10.9017	Adder	12.83
290108	LEEDK;1U E	25.2161	Adder	29.67
293061	N-015 E	16.4567	Adder	19.36
293644	O22 E1	10.6010	Adder	12.47
293645	O22 E2	20.5784	Adder	24.21
294392	P-010 E	20.9000	Adder	24.59
910542	X3-005 E	0.7569	Adder	0.89
915011	Y3-013 1	3.8668	Adder	4.55
915021	Y3-013 2	3.8668	Adder	4.55
915031	Y3-013 3	3.8668	Adder	4.55
916221	Z1-073 E	5.6251	Adder	6.62
917502	Z2-087 E	21.5594	Adder	25.36
918052	AA1-018 E OP	16.9981	Adder	20.0
920272	AA2-123 E	2.5284	Adder	2.97

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
924041	AB2-047 C O1	4.0269	Adder	4.74
924042	AB2-047 E O1	26.9492	Adder	31.7
924471	AB2-096	43.8128	Adder	51.54
925302	AB2-191 E	1.4436	Adder	1.7
926311	AC1-109 1	1.9783	Adder	2.33
926321	AC1-109 2	1.9783	Adder	2.33
926331	AC1-110 1	1.9652	Adder	2.31
926341	AC1-110 2	1.9652	Adder	2.31
926351	AC1-111 1	0.7901	Adder	0.93
926361	AC1-111 2	0.7901	Adder	0.93
926371	AC1-111 3	0.7901	Adder	0.93
926381	AC1-111 4	0.7901	Adder	0.93
926391	AC1-111 5	0.7901	Adder	0.93
926401	AC1-111 6	0.7901	Adder	0.93
927451	AC1-142A 1	4.3292	Adder	5.09
927461	AC1-142A 2	4.3292	Adder	5.09
930481	AB1-089	68.0633	Adder	80.07
930501	AB1-091 O1	79.4049	Adder	93.42
930741	AB1-122 1O1	75.3653	Adder	88.67
930751	AB1-122 2O1	76.2988	Adder	89.76
932881	AC2-115 1	0.7443	Adder	0.88
932891	AC2-115 2	0.7443	Adder	0.88
932921	AC2-116	0.8619	Adder	1.01
933411	AC2-154 C	2.7084	Adder	3.19
933412	AC2-154 E	4.4189	Adder	5.2
933431	AC2-156 C O1	0.9902	Adder	1.16
933432	AC2-156 E O1	1.6156	Adder	1.9
933911	AD1-013 C	1.9249	Adder	2.26
933912	AD1-013 E	3.0748	Adder	3.62
933931	AD1-016 C	0.9608	Adder	1.13
933932	AD1-016 E	1.5676	Adder	1.84
934101	AD1-039 1	7.3858	Adder	8.69
934111	AD1-039 2	7.4773	Adder	8.8
934431	AD1-067 C	0.1369	Adder	0.16
934432	AD1-067 E	0.5756	Adder	0.68
934701	AD1-098 C O1	7.1725	Adder	8.44
934702	AD1-098 E O1	5.2367	Adder	6.16
934721	AD1-100 C	24.8701	Adder	29.26
934722	AD1-100 E	116.0607	Adder	136.54
934871	AD1-116 C	0.9899	Adder	1.16
934872	AD1-116 E	1.6151	Adder	1.9
934971	AD1-129 C	0.9353	Adder	1.1
934972	AD1-129 E	0.6236	Adder	0.73
935001	AD1-133 C O1	23.0433	Adder	27.11
935002	AD1-133 E O1	15.3622	Adder	18.07
936291	AD2-038 C O1	2.4322	Adder	2.86
936292	AD2-038 E O1	16.2771	Adder	19.15
936371	AD2-047 C O1	4.8465	Adder	5.7
936372	AD2-047 E O1	23.6625	Adder	27.84
936461	AD2-060	2.8509	Adder	3.35
936511	AD2-066 C O1	8.7722	Adder	10.32
936512	AD2-066 E O1	5.8482	Adder	6.88

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
937001	AD2-134 C	2.8499	Adder	3.35
937002	AD2-134 E	11.7728	Adder	13.85
937401	AD2-194 1	8.0764	Adder	9.5
937411	AD2-194 2	8.0764	Adder	9.5
938511	AE1-070 1	9.4897	Adder	11.16
938521	AE1-070 2	8.6821	Adder	10.21
938851	AE1-113 C	8.2686	Adder	9.73
938852	AE1-113 E	29.3159	Adder	34.49
939321	AE1-163 C O1	6.1117	Adder	7.19
939322	AE1-163 E O1	37.5434	Adder	44.17
939351	AE1-166 C O1	12.2715	Adder	14.44
939352	AE1-166 E O1	11.3275	Adder	13.33
939401	AE1-172 C O1	6.7996	Adder	8.0
939402	AE1-172 E O1	31.9033	Adder	37.53
939741	AE1-205 C O1	10.4080	Adder	12.24
939742	AE1-205 E O1	14.3729	Adder	16.91
940101	AE1-252 C O1	13.6598	Adder	16.07
940102	AE1-252 E O1	9.1066	Adder	10.71
940752	AE2-062 E	0.1352	Adder	0.16
941131	AE2-107 C	7.5832	Adder	8.92
941132	AE2-107 E	5.0555	Adder	5.95
941551	AE2-152 C O1	14.1594	Adder	16.66
941552	AE2-152 E O1	9.4396	Adder	11.11
941561	AE2-153 C O1	5.1289	Adder	6.03
941562	AE2-153 E O1	24.0125	Adder	28.25
941731	AE2-173 O1	6.1952	Adder	7.29
942111	AE2-223 C	2.4161	Adder	2.84
942112	AE2-223 E	16.1695	Adder	19.02
942421	AE2-255 C O1	3.1320	Adder	3.68
942422	AE2-255 E O1	9.3961	Adder	11.05
942651	AE2-281 C O1	0.8731	Adder	1.03
942652	AE2-281 E O1	5.3633	Adder	6.31
942991	AE2-321 C	8.4588	Adder	9.95
942992	AE2-321 E	4.1663	Adder	4.9
943121	AE2-341 C	13.2198	Adder	15.55
943122	AE2-341 E	6.4917	Adder	7.64
943381	AF1-009 C	0.6205	Adder	0.73
943382	AF1-009 E	2.4818	Adder	2.92
943591	AF1-030 C O1	8.7913	Adder	10.34
943592	AF1-030 E O1	4.3497	Adder	5.12
943801	AF1-048 C	3.9542	Adder	4.65
943802	AF1-048 E	2.6361	Adder	3.1
944041	AF1-072	2.2279	Adder	2.62
944911	AF1-156 C	13.1136	Adder	15.43
944912	AF1-156 E	8.7424	Adder	10.29
945351	AF1-200 FTIR	322.9138	Merchant Transmission	322.9138
946661	AF1-330 C	2.0733	Adder	2.44
946662	AF1-330 E	0.4551	Adder	0.54
946671	AF1-331	2.4998	Adder	2.94
WEC	WEC	4.5647	Confirmed LTF	4.5647
CBM-W2	CBM-W2	25.2006	Confirmed LTF	25.2006
NY	NY	1.4251	Confirmed LTF	1.4251

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
TVA	TVA	1.6366	Confirmed LTF	1.6366
O-066	O-066	16.9008	Confirmed LTF	16.9008
CHEOAH	CHEOAH	0.0711	Confirmed LTF	0.0711
CBM-S1	CBM-S1	6.1514	Confirmed LTF	6.1514
G-007	G-007	2.6125	Confirmed LTF	2.6125
MADISON	MADISON	24.3210	Confirmed LTF	24.3210
MEC	MEC	16.0378	Confirmed LTF	16.0378
CALDERWOOD	CALDERWOOD	0.0567	Confirmed LTF	0.0567
BLUEG	BLUEG	2.2846	Confirmed LTF	2.2846
TRIMBLE	TRIMBLE	0.7897	Confirmed LTF	0.7897
CATAWBA	CATAWBA	0.3577	Confirmed LTF	0.3577
CBM-W1	CBM-W1	78.3626	Confirmed LTF	78.3626

11.7.5 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
44099381	275233	WILTON ;4M	CE	270644	WILTON ;	CE	1	COMED_P4_112-65-BT2-3__	breaker	1379.0	117.77	117.94	AC	15.79

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
274771	LINCOLN ;2U	3.9737	50/50	3.9737
274772	LINCOLN ;3U	3.9737	50/50	3.9737
274773	LINCOLN ;4U	3.9737	50/50	3.9737
274774	LINCOLN ;5U	3.9737	50/50	3.9737
274775	LINCOLN ;6U	3.9737	50/50	3.9737
274776	LINCOLN ;7U	3.9737	50/50	3.9737
274777	LINCOLN ;8U	3.9737	50/50	3.9737
274830	U3-021 1	6.4436	Adder	7.58
274831	U3-021 2	6.4436	Adder	7.58
274881	PILOT HIL;1E	20.3728	Adder	23.97
274890	CAYUG;1U E	17.3594	Adder	20.42
274891	CAYUG;2U E	17.3594	Adder	20.42
275149	KELLYCK ;1E	20.3728	Adder	23.97
276167	Z1-106 E2	1.3354	Adder	1.57
276168	Z1-106 E1	1.3353	Adder	1.57
276169	Z1-107 E	2.7333	Adder	3.22
276170	Z1-108 E	2.6275	Adder	3.09
290021	O50 E	20.4721	Adder	24.08
290051	GSG-6; E	11.1333	Adder	13.1
290108	LEEDK;1U E	25.7533	Adder	30.3
293061	N-015 E	16.8004	Adder	19.77
293644	O22 E1	10.8250	Adder	12.74
293645	O22 E2	21.0133	Adder	24.72
294392	P-010 E	21.3365	Adder	25.1
910542	X3-005 E	0.7734	Adder	0.91
915011	Y3-013 1	3.9494	Adder	4.65
915021	Y3-013 2	3.9494	Adder	4.65

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
915031	Y3-013 3	3.9494	Adder	4.65
916221	Z1-073 E	5.7446	Adder	6.76
917502	Z2-087 E	21.9972	Adder	25.88
918052	AA1-018 E OP	17.3697	Adder	20.43
920272	AA2-123 E	2.5823	Adder	3.04
924041	AB2-047 C O1	4.1087	Adder	4.83
924042	AB2-047 E O1	27.4965	Adder	32.35
924471	AB2-096	44.7470	Adder	52.64
925302	AB2-191 E	1.4743	Adder	1.73
926311	AC1-109 1	2.0204	Adder	2.38
926321	AC1-109 2	2.0204	Adder	2.38
926331	AC1-110 1	2.0070	Adder	2.36
926341	AC1-110 2	2.0070	Adder	2.36
926351	AC1-111 1	0.8070	Adder	0.95
926361	AC1-111 2	0.8070	Adder	0.95
926371	AC1-111 3	0.8070	Adder	0.95
926381	AC1-111 4	0.8070	Adder	0.95
926391	AC1-111 5	0.8070	Adder	0.95
926401	AC1-111 6	0.8070	Adder	0.95
927451	AC1-142A 1	4.4224	Adder	5.2
927461	AC1-142A 2	4.4224	Adder	5.2
930481	AB1-089	69.5172	Adder	81.78
930501	AB1-091 O1	81.0552	Adder	95.36
930741	AB1-122 1O1	76.9684	Adder	90.55
930751	AB1-122 2O1	77.9263	Adder	91.68
932881	AC2-115 1	0.7601	Adder	0.89
932891	AC2-115 2	0.7601	Adder	0.89
932921	AC2-116	0.8802	Adder	1.04
933411	AC2-154 C	2.7647	Adder	3.25
933412	AC2-154 E	4.5109	Adder	5.31
933431	AC2-156 C O1	1.0112	Adder	1.19
933432	AC2-156 E O1	1.6499	Adder	1.94
933911	AD1-013 C	1.9659	Adder	2.31
933912	AD1-013 E	3.1403	Adder	3.69
933931	AD1-016 C	0.9813	Adder	1.15
933932	AD1-016 E	1.6010	Adder	1.88
934101	AD1-039 1	7.5429	Adder	8.87
934111	AD1-039 2	7.6368	Adder	8.98
934431	AD1-067 C	0.1398	Adder	0.16
934432	AD1-067 E	0.5878	Adder	0.69
934701	AD1-098 C O1	7.3253	Adder	8.62
934702	AD1-098 E O1	5.3482	Adder	6.29
934721	AD1-100 C	25.3636	Adder	29.84
934722	AD1-100 E	118.3633	Adder	139.25
934871	AD1-116 C	1.0116	Adder	1.19
934872	AD1-116 E	1.6505	Adder	1.94
934971	AD1-129 C	0.9552	Adder	1.12
934972	AD1-129 E	0.6368	Adder	0.75
935001	AD1-133 C O1	23.5222	Adder	27.67
935002	AD1-133 E O1	15.6815	Adder	18.45
936291	AD2-038 C O1	2.4848	Adder	2.92
936292	AD2-038 E O1	16.6288	Adder	19.56

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
936371	AD2-047 C O1	4.9474	Adder	5.82
936372	AD2-047 E O1	24.1549	Adder	28.42
936461	AD2-060	2.9102	Adder	3.42
936511	AD2-066 C O1	8.9580	Adder	10.54
936512	AD2-066 E O1	5.9720	Adder	7.03
937001	AD2-134 C	2.9104	Adder	3.42
937002	AD2-134 E	12.0229	Adder	14.14
937401	AD2-194 1	8.2498	Adder	9.71
937411	AD2-194 2	8.2493	Adder	9.71
938511	AE1-070 1	9.6935	Adder	11.4
938521	AE1-070 2	8.8679	Adder	10.43
938851	AE1-113 C	8.4447	Adder	9.93
938852	AE1-113 E	29.9404	Adder	35.22
939321	AE1-163 C O1	6.2438	Adder	7.35
939322	AE1-163 E O1	38.3545	Adder	45.12
939351	AE1-166 C O1	12.5188	Adder	14.73
939352	AE1-166 E O1	11.5558	Adder	13.6
939401	AE1-172 C O1	6.9344	Adder	8.16
939402	AE1-172 E O1	32.5358	Adder	38.28
939741	AE1-205 C O1	10.6193	Adder	12.49
939742	AE1-205 E O1	14.6648	Adder	17.25
940101	AE1-252 C O1	13.9306	Adder	16.39
940102	AE1-252 E O1	9.2871	Adder	10.93
940752	AE2-062 E	0.1381	Adder	0.16
941131	AE2-107 C	7.7444	Adder	9.11
941132	AE2-107 E	5.1629	Adder	6.07
941551	AE2-152 C O1	14.4447	Adder	16.99
941552	AE2-152 E O1	9.6298	Adder	11.33
941561	AE2-153 C O1	5.2345	Adder	6.16
941562	AE2-153 E O1	24.5070	Adder	28.83
941731	AE2-173 O1	6.3210	Adder	7.44
942111	AE2-223 C	2.4652	Adder	2.9
942112	AE2-223 E	16.4979	Adder	19.41
942421	AE2-255 C O1	3.1988	Adder	3.76
942422	AE2-255 E O1	9.5963	Adder	11.29
942651	AE2-281 C O1	0.8920	Adder	1.05
942652	AE2-281 E O1	5.4792	Adder	6.45
942991	AE2-321 C	8.6393	Adder	10.16
942992	AE2-321 E	4.2552	Adder	5.01
943121	AE2-341 C	13.5012	Adder	15.88
943122	AE2-341 E	6.6298	Adder	7.8
943381	AF1-009 C	0.6337	Adder	0.75
943382	AF1-009 E	2.5347	Adder	2.98
943591	AF1-030 C O1	8.9784	Adder	10.56
943592	AF1-030 E O1	4.4422	Adder	5.23
943801	AF1-048 C	4.0386	Adder	4.75
943802	AF1-048 E	2.6924	Adder	3.17
944041	AF1-072	2.2755	Adder	2.68
944911	AF1-156 C	13.3837	Adder	15.75
944912	AF1-156 E	8.9225	Adder	10.5
945351	AF1-200 FTIR	329.7310	Merchant Transmission	329.7310
946661	AF1-330 C	2.1175	Adder	2.49

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
946662	AF1-330 E	0.4648	Adder	0.55
946671	AF1-331	2.5531	Adder	3.0
WEC	WEC	4.6620	Confirmed LTF	4.6620
CBM-W2	CBM-W2	25.7248	Confirmed LTF	25.7248
NY	NY	1.4555	Confirmed LTF	1.4555
TVA	TVA	1.6688	Confirmed LTF	1.6688
O-066	O-066	17.2637	Confirmed LTF	17.2637
CHEOAH	CHEOAH	0.0731	Confirmed LTF	0.0731
CBM-S1	CBM-S1	6.2707	Confirmed LTF	6.2707
G-007	G-007	2.6686	Confirmed LTF	2.6686
MADISON	MADISON	24.8371	Confirmed LTF	24.8371
MEC	MEC	16.3762	Confirmed LTF	16.3762
CALDERWOOD	CALDERWOOD	0.0586	Confirmed LTF	0.0586
BLUEG	BLUEG	2.3349	Confirmed LTF	2.3349
TRIMBLE	TRIMBLE	0.8069	Confirmed LTF	0.8069
CATAWBA	CATAWBA	0.3654	Confirmed LTF	0.3654
CBM-W1	CBM-W1	80.0390	Confirmed LTF	80.0390

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA1-018	Powerton-Goodings Grove	In Service
AA2-030	Nelson	Active
AA2-123	Marengo 34kV	In Service
AB1-089	Byron-Wayne 345kV #1	Active
AB1-091	Davis Creek 345kV	Active
AB1-122	Kendall-Tazewell & Dresden-Mole Creek	Active
AB2-047	Brokaw-Pontiac Midpoint	Engineering and Procurement
AB2-096	Silver Lake-Cherry Valley	Active
AB2-191	Mendota Hills	In Service
AC1-033	Kewanee	Active
AC1-109	Aurora 345kV	Active
AC1-110	Aurora 138kV	Active
AC1-111	Aurora 138kV	Active
AC1-113	Rockford	Active
AC1-114	Rockford II	Active
AC1-142A	Joliet	Active
AC1-168	Kewanee-Streator	Active
AC1-214	Crescent Ridge	Engineering and Procurement
AC2-115	Rockford	Active

Queue Number	Project Name	Status
AC2-116	Rockford II	Active
AC2-117	University Park North	Engineering and Procurement
AC2-154	Davis Creek 138kV	
AC2-156	Sandwich 34.5kV	
AD1-013	Twombly Road 138kV	Active
AD1-016	Marengo	Active
AD1-031	Kewanee 138 kV	Active
AD1-039	Kendall-Tazewell & Dresden-Mole Creek	Active
AD1-067	Mendota Hills	Active
AD1-096	Stillman Valley 34 kV	Active
AD1-098	Dixon-McGirr	Active
AD1-100	Loretto-Wilton & Braidwood-Davis Creek	Active
AD1-116	Nevada 345 kV	Active
AD1-129	Belvidere 34 kV	Active
AD1-133	Pontiac MidPoint-Dresden	Active
AD2-038	Powerton	Active
AD2-047	Davis Creek 138 kV	Active
AD2-060	Davis Creek 138kV	Active
AD2-066	Mazon-Crescent Ridge	Active
AD2-102	Wempletown 345 kV	Active
AD2-134	Shady Oaks	Active
AD2-172	Lena 138kV	Active
AD2-194	Elwood	Active
AD2-214	Rock Falls-Garden Plains	Active
AE1-070	Elwood 345 kV	Active
AE1-113	Mole Creek 345 kV	Active
AE1-114	Maryland-Lancaster 138 kV	Active
AE1-134	Nelson 345 kV	Active
AE1-163	Powerton-Nevada 345 kV	Active
AE1-166	Loretto-Wilton & Braidwood-Davis Creek	Active
AE1-172	Loretto-Wilton Center	Active
AE1-205	McLean 345 kV	Active
AE1-252	Loretto-Wilton Center	Active
AE2-035	Lena 138 kV	Active
AE2-062	Normantown	Active
AE2-107	Haumesser Road 138 kV	Active
AE2-152	Loretto-Wilton & Braidwood-Davis Creek	Active
AE2-153	Braidwood-Davis Creek	Active
AE2-173	McLean 345 kV	Active
AE2-223	McLean 345 kV	Active
AE2-255	Molecreek 345 kV	Active
AE2-281	Powerton-Nevada 345 kV	Active
AE2-321	Belvidere-Marengo 138 kV	Active
AE2-341	Sandwich-Plano	Active
AF1-009	Dixon-McGirr	Active
AF1-011	Schauff Road	Active
AF1-012	Electric Junc-Nelson	Active
AF1-030	Plano-R 138 kV	Active
AF1-048	Belvidere-Marengo	Active
AF1-060	Lena 138 kV	Active
AF1-072	Rocky Road	Active
AF1-156	Braidwood-Davis Creek	Active

Queue Number	Project Name	Status
AF1-200	Plano 345 kV	Active
AF1-280	Nelson-Lee County	Active
AF1-281	Nelson-Lee County	Active
AF1-296	Garden Plain 138 kV	Active
AF1-314	Lena 138 kV	Active
AF1-317	Electric Jct-Nelson	Active
AF1-318	Crescent Ridge-Corbin	Active
AF1-330	Marengo	Active
AF1-331	Twombly Road	Active
U3-021	Silver Lake-Cherry Valley	Active
X3-005	Wildwood 12kV	In Service
Y3-013	Zion Energy Center	Active
Z1-072	Crescent Ridge	In Service
Z1-073	Mendota Hills	In Service
Z1-106	West Chicago 34kV	In Service
Z1-107	Joliet 34kV	In Service
Z1-108	McHenry 34kV	In Service
Z2-081	Streator 34.5kV	In Service
Z2-087	Pontiac MidPoint-Brokaw 345kV	In Service

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
COMED_P4_167-45-BT5-6__	CONTINGENCY 'COMED_P4_167-45-BT5-6__' TRIP BRANCH FROM BUS 270846 TO BUS 270847 CKT 1 / PLANO ; B 345 PLANO ; R 345 TRIP BRANCH FROM BUS 270847 TO BUS 270733 CKT 1 / PLANO ; R 345 ELEC JUNC;3R 345 TRIP BRANCH FROM BUS 275208 TO BUS 270630 CKT 1 / PLANO ;4M 345 PLANO ; 765 TRIP BRANCH FROM BUS 275208 TO BUS 270847 CKT 1 / PLANO ;4M 345 PLANO ; R 345 TRIP BRANCH FROM BUS 275208 TO BUS 275308 CKT 1 / PLANO ;4M 345 PLANO ;4C 33 END
COMED_P4_112-65-BT5-6__	CONTINGENCY 'COMED_P4_112-65-BT5-6__' TRIP BRANCH FROM BUS 270644 TO BUS 270607 CKT 1 / WILTO; 765 COLLI; 765 TRIP BRANCH FROM BUS 275233 TO BUS 270644 CKT 1 / WILTO;4M 345 WILTO; 765 TRIP BRANCH FROM BUS 275233 TO BUS 270927 CKT 1 / WILTO;4M 345 WILTO; R 345 TRIP BRANCH FROM BUS 275233 TO BUS 275333 CKT 1 / WILTO;4M 345 WILTO;4C 33 END

Contingency Name	Contingency Definition
AEP_P4_#2978_05DUMONT_FSA	CONTINGENCY 'AEP_P4_#2978_05DUMONT_FSA' OPEN BRANCH FROM BUS 243206 TO BUS 907040 CKT 1 / 243206 05DUMONT 765 X1-020 OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206 05DUMONT 765 270644 WILTON ; 765 1 END
COMED_P4_112-65-BT2-3__	CONTINGENCY 'COMED_P4_112-65-BT2-3__' TRIP BRANCH FROM BUS 270644 TO BUS 270607 CKT 1 / WILTO; 765 COLLI; 765 TRIP BRANCH FROM BUS 275232 TO BUS 270644 CKT 1 / WILTO;3M 345 WILTO; 765 TRIP BRANCH FROM BUS 275232 TO BUS 270926 CKT 1 / WILTO;3M 345 WILTO; B 345 TRIP BRANCH FROM BUS 275232 TO BUS 275332 CKT 1 / WILTO;3M 345 WILTO;3C 33 END
COMED_P1-2_695_B2	CONTINGENCY 'COMED_P1-2_695_B2' OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206 05DUMONT 765 270644 WILTO; 765 1 END
COMED_P4_111-38-TR82__	CONTINGENCY 'COMED_P4_111-38-TR82__' TRIP BRANCH FROM BUS 271390 TO BUS 271586 CKT 1 / ELECT; B 138 W541 ; B 138 TRIP BRANCH FROM BUS 271390 TO BUS 272724 CKT 1 / ELECT; B 138 WARRE;BT 138 TRIP BRANCH FROM BUS 271390 TO BUS 275239 CKT 1 / ELECT; B 138 ELECT;2M 138 MOVE 100 PERCENT LOAD FROM BUS 271586 TO BUS 271587 / W541 ; B 138 W541 ; R 138 MOVE 100 PERCENT LOAD FROM BUS 272522 TO BUS 272523 / SUGAR; B 138 SUGAR; R 138 CLOSE LINE FROM BUS 272114 TO BUS 272115 CKT 1 / N AUR; B 138 N AUR; R 138 DISCONNECT BUS 271560 / GLIDD;BT 138 DISCONNECT BUS 272522 / SUGAR; B 138 DISCONNECT BUS 275239 / ELECT;2M 138 REMOVE SWSHUNT FROM BUS 271390 END
272803 W PLANO ; R 138 943120 AE2-341 TAP 138 1	CONTINGENCY '272803 W PLANO ; R 138 943120 AE2-341 TAP 138 1' OPEN BRANCH FROM BUS 272803 TO BUS 943120 CKT 1 END
COMED_P4_112-65-BT3-4__	CONTINGENCY 'COMED_P4_112-65-BT3-4__' TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765 05DUMONT 765 TRIP BRANCH FROM BUS 275232 TO BUS 270644 CKT 1 / WILTO;3M 345 WILTO; 765 TRIP BRANCH FROM BUS 275232 TO BUS 270926 CKT 1 / WILTO;3M 345 WILTO; B 345 TRIP BRANCH FROM BUS 275232 TO BUS 275332 CKT 1 / WILTO;3M 345 WILTO;3C 33 END

Contingency Name	Contingency Definition
COMED_P1-2_138-L11106_B-R	CONTINGENCY 'COMED_P1-2_138-L11106_B-R' TRIP BRANCH FROM BUS 271390 TO BUS 271586 CKT 1 / ELECT; B 138 W541 ; B 138 TRIP BRANCH FROM BUS 271560 TO BUS 271558 CKT 1 / GLIDD;BT 138 GLIDD; B 138 TRIP BRANCH FROM BUS 271560 TO BUS 272728 CKT 1 / GLIDD;BT 138 WATER; B 138 TRIP BRANCH FROM BUS 271586 TO BUS 272114 CKT 1 / W541 ; B 138 N AUR; B 138 TRIP BRANCH FROM BUS 272114 TO BUS 272522 CKT 1 / N AUR; B 138 SUGAR; B 138 TRIP BRANCH FROM BUS 272522 TO BUS 271560 CKT 1 / SUGAR; B 138 GLIDD;BT 138 MOVE 100 PERCENT LOAD FROM BUS 271586 TO BUS 271587 / W541 ; B 138 W541 ; R 138 MOVE 100 PERCENT LOAD FROM BUS 272522 TO BUS 272523 / SUGAR; B 138 SUGAR; R 138 CLOSE LINE FROM BUS 272114 TO BUS 272115 CKT 1 / N AUR; B 138 N AUR; R 138 END
Base Case	
COMED_P4_023-65-BT2-3__	CONTINGENCY 'COMED_P4_023-65-BT2-3__' TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765 05DUMONT 765 TRIP BRANCH FROM BUS 270607 TO BUS 270630 CKT 1 / COLLI; 765 PLANO; 765 END
COMED_P4_083-38-BT3-4__	CONTINGENCY 'COMED_P4_083-38-BT3-4__' TRIP BRANCH FROM BUS 271390 TO BUS 271586 CKT 1 / ELECT; B 138 W541 ; B 138 TRIP BRANCH FROM BUS 271560 TO BUS 271558 CKT 1 / GLIDD;BT 138 GLIDD; B 138 TRIP BRANCH FROM BUS 271560 TO BUS 272728 CKT 1 / GLIDD;BT 138 WATER; B 138 TRIP BRANCH FROM BUS 271586 TO BUS 272114 CKT 1 / W541 ; B 138 N AUR; B 138 TRIP BRANCH FROM BUS 272114 TO BUS 272522 CKT 1 / N AUR; B 138 SUGAR; B 138 TRIP BRANCH FROM BUS 272522 TO BUS 271560 CKT 1 / SUGAR; B 138 GLIDD;BT 138 MOVE 100 PERCENT LOAD FROM BUS 271586 TO BUS 271587 / W541 ; B 138 W541 ; R 138 MOVE 100 PERCENT LOAD FROM BUS 272522 TO BUS 272523 / SUGAR; B 138 SUGAR; R 138 CLOSE LINE FROM BUS 272114 TO BUS 272115 CKT 1 / N AUR; B 138 N AUR; R 138 TRIP BRANCH FROM BUS 271558 TO BUS 272730 CKT 1 / GLIDD; B 138 WATER;3B 138 MOVE 100 PERCENT LOAD FROM BUS 272761 TO BUS 272759 / W DEK;7R 138 W DEK;4R 138 DISCONNECT BUS 271581 / B200 ; R 138 DISCONNECT BUS 272757 / W DEK;7T 138 END

Contingency Name	Contingency Definition
AEP_P4_#2978_05DUMONT_NON_FSA	CONTINGENCY 'AEP_P4_#2978_05DUMONT_NON_FSA' OPEN BRANCH FROM BUS 243206 TO BUS 907040 CKT 1 / 243206 05DUMONT 765 X1-020 OPEN BRANCH FROM BUS 243207 TO BUS 907040 CKT 1 OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206 05DUMONT 765 270644 WILTON ; 765 1 END
COMED_P4_112-65-BT4-5__	CONTINGENCY 'COMED_P4_112-65-BT4-5__' TRIP BRANCH FROM BUS 270644 TO BUS 243206 CKT 1 / WILTO; 765 05DUMONT 765 TRIP BRANCH FROM BUS 275233 TO BUS 270644 CKT 1 / WILTO;4M 345 WILTO; 765 TRIP BRANCH FROM BUS 275233 TO BUS 270927 CKT 1 / WILTO;4M 345 WILTO; R 345 TRIP BRANCH FROM BUS 275233 TO BUS 275333 CKT 1 / WILTO;4M 345 WILTO;4C 33 END

12 Light Load Analysis

Not Required

13 Short Circuit Analysis

The following breakers are overdutied:

None

13.1 System Reinforcements - Short Circuit

None

14 Stability and Reactive Power

To be determined in the Facilities Study Phase.

15 Affected Systems

15.1 MISO

MISO Impacts to be determined during later study phases (as applicable).

16 Attachment 1: One Line Diagram

